

Multiple Choice

1. **Answer:** B

Options: A) It is a colorless gas.; B) It is a red-brown volatile liquid.; C) It is a colorless volatile liquid.; D) It is a yellow metallic solid.

Note: Correct option: B - It is a red-brown volatile liquid.

2. **Answer:** A

Options: A) It has little effect.; B) It significantly increases the pH.; C) It significantly decreases the pH.; D) It changes the pH asymptotically to the pKa of the acid.

Note: Correct option: A - It has little effect.

3. **Answer:** C

Options: A) $101.1 \times 10^7 \text{ T}^{-1} \text{ s}^{-1}$; B) $26.75 \times 10^7 \text{ T}^{-1} \text{ s}^{-1}$; C) $7.081 \times 10^7 \text{ T}^{-1} \text{ s}^{-1}$; D) $0.9377 \times 10^7 \text{ T}^{-1} \text{ s}^{-1}$

Note: Correct option: C - $7.081 \times 10^7 \text{ T}^{-1} \text{ s}^{-1}$

4. **Answer:** A

Options: A) carbonic acid; B) a halogenated hydrocarbon; C) lead; D) mercury

Note: Correct option: A - carbonic acid

5. **Answer:** D

Options: A) Ammonium hydroxide; B) Sulfuric acid; C) Acetic acid; D) Potassium hydrogen phthalate

Note: Correct option: D - Potassium hydrogen phthalate

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'trigonal pyramidal'.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: ' B_{12} icosahedra'.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** 4.62 ppm. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** NaCl. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key